Concentration Inequality

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Definitions and theorems are adapted from Karthik Sridharan's wonderful article:
https://www.cs.cornell.edu/~sridharan/concentration.pdf

Recap: Problem and Motivation

- Evaluate $P(Z|X)$ and related expectations

- Problem with exact methods
  - Curse of dimensionality

  - $P(Z|X)$ has a complex form making expectations analytically intractable

Recap: Variational Inference

- Functional: a function that maps a function to a value

$$\mathrm{H}[p] = \int p(x) \ln p(x) \, \mathrm{d}x$$

- Variational method: find an input function that maximizes the functional

- Variational inference: find a distribution q(z) to approximate p(Z|X) so a functional is maximized

Recap: Variational Inference

$$\ln p(\mathbf{X}) = \mathcal{L}(q) + \mathrm{KL}(q\|p)$$

Between p(Z|X) and q(Z)

$$\mathcal{L}(q) = \int q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z})} \right\} \mathrm{d}\mathbf{Z}$$

$$\mathrm{KL}(q\|p) = -\int q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X})}{q(\mathbf{Z})} \right\} \mathrm{d}\mathbf{Z}$$

If q can be any distribution, then variational inference is precise.
But in practice, it cannot

Is the following statement right?

- Probability $p(Z,X)$ is usually easier to evaluate compared to $P(Z|X)$.

Recap: Sampling Methods

- Stochastic methods

- Also called Monte Carlo methods

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})\,\mathrm{d}\mathbf{z} \quad \Rightarrow \quad \hat{f} = \frac{1}{L}\sum_{l=1}^{L} f(\mathbf{z}^{(l)})$$

$\mathbf{z}_1, \ldots, \mathbf{z}_l$ are samples from p

Recap: Sampling Methods

- Transformation method: CDF^{-1}(uniform(0,1))

- Rejection sampling
  - A proposal distribution q(z)
  - Choose k, such that k*q(z) >= p(z), for any x
  - Sampling process:
    - Sample z_0 from q(z)
    - Sample h from uniform(0, k*q(z_0))
    - If h > p(z_0), discard it; otherwise, keep it

Is the following statement correct?

- All primitive distributions can be constructed using the inverse CDF.

Is the following statement right?

- In rejection sampling, given k, the probability whether a sample is accepted does not depend on the proposal distribution

Is the following statement correct?

- The efficiency of rejection sampling depends on the choice of the proposal distribution

Recap: Sampling Methods

- Importance sampling
  - Used to evaluate f(z) where z is from p(z)

$$E(f) = \int f(z)p(z)dz = \int f(z)\frac{p(z)}{q(z)}q(z)dz \approx \frac{1}{L}\sum_{l=1}^{L}\frac{p(z^l)}{q(z^l)}f(z^l)$$

  - How to get real samples: create a new discrete distribution using the above samples and set their probabilities using the importance weights

Recap: Sampling Methods

- Markov Chain Monte Carlo
  - A sampling method that works with a large family of distributions and high dimensions

- Workflow
  - Start with some sample z_0
  - Suppose the current sample is z^τ . Draw next sample z^* from $q(z \mid z^\tau)$
  - Decide whether to accept z^*as the next state based some criteria. If accepted, $z^{\tau+1} = z^*$. Otherwise, $z^{\tau+1} = z^\tau$
  - Samples form a Markov chain

Recap: Sampling Methods

	Metropolis	Metropolis-Hasting
Constraints on the proposal distribution	Symmetric	None
Accepting probability	$\min(1, \frac{p(z')}{p(z)})$	$\min(1, \frac{p(z')q(z\|z')}{p(z)q(z'\|z)})$

Recap: Why MCMC works?

- Markov chain: $$p(\mathbf{z}^{(m+1)}|\mathbf{z}^{(1)},\ldots,\mathbf{z}^{(m)}) = p(\mathbf{z}^{(m+1)}|\mathbf{z}^{(m)}).$$

- Stationary distribution of a Markov chain: each step in the chain does not change the distribution.

- Detailed balance: $$p^\star(\mathbf{z})T(\mathbf{z},\mathbf{z}') = p^\star(\mathbf{z}')T(\mathbf{z}',\mathbf{z})$$
  - $p^*(\mathbf{z})$ is a stationary distribution

- A *ergodic* Markov chain converges to the same distribution regardless the initial distribution
  - The system does not return to the same state at fixed intervals
  - The expected number of steps for returning to the same state is finite

Is the following statement right?

- The samples drawn using MCMC are independent

Is the following statement right?

- A Markov chain can have more than one stationary distribution

Use MCMC to solve the problem below

- Super optimization
  - There is a straight-line program
  - A set of test cases are given
  - The program can be modified by deleting a statement, inserting a statement from the initial program at a given place
  - Optimize the program by using the above operations

Outline

- Introduction
- Related notations
- Instances

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Motivations

- I want to estimate the expectation of a random variables. I have taken 1000 samples and calculated the mean. How accurate is the estimation?

Definition

- "In probability theory, concentration inequalities provide mathematical bounds on the probability of a random variable deviating from some value (typically, its expected value). " – Wikipedia

- Given a random variable Y and a constant m, Y is said to be **concentrated** around m if,

$$P(|Y - m| > t) \leq D(t)$$

where, $D(t)$ decreases drastically to 0 in t.

Source: https://www.epfl.ch/labs/lions/wp-content/uploads/2019/01/Scribe-Lecture-10.pdf

Examples

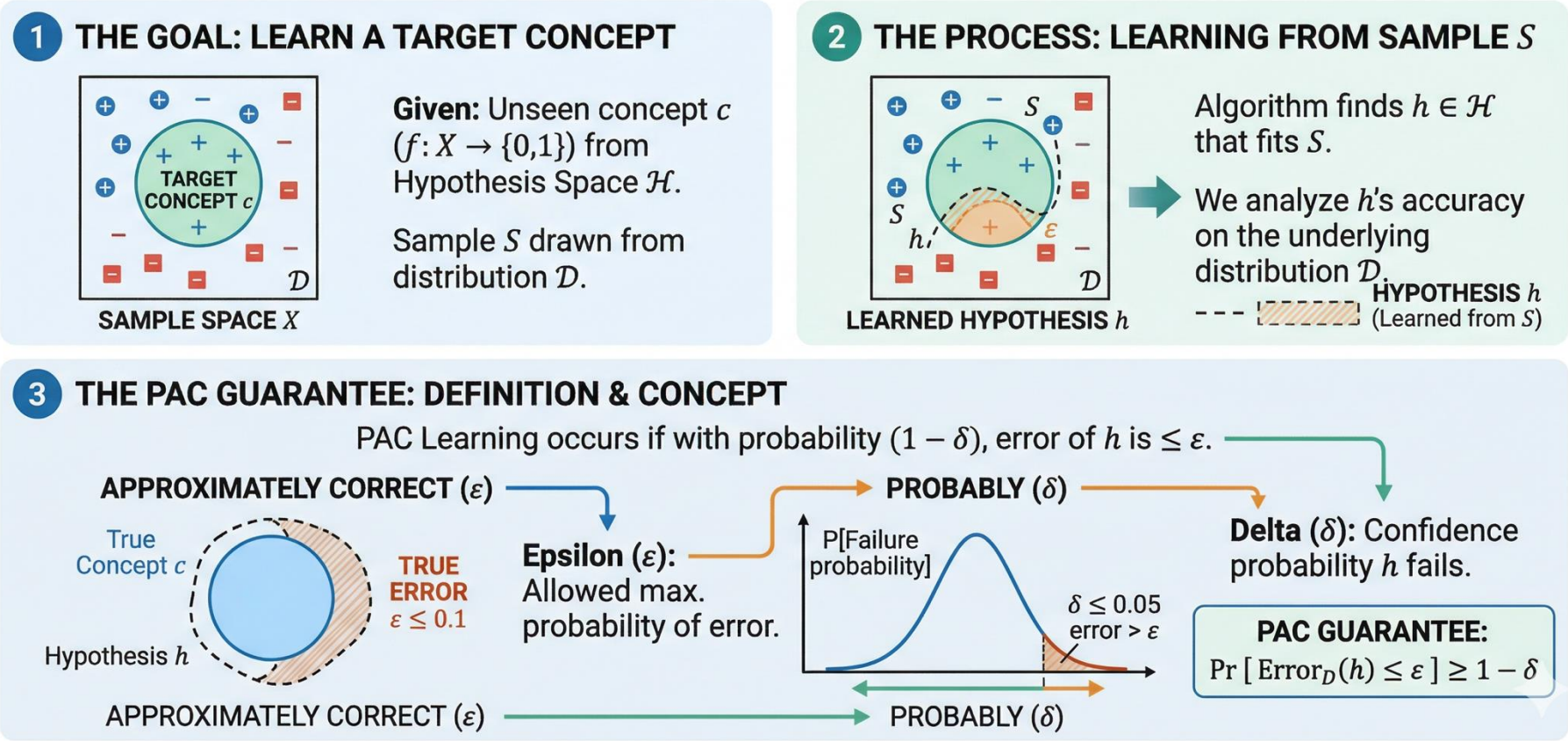
- Law of large numbers: Let $Y = \frac{1}{n}\sum_{i=1}^{n} X_i$, where X_i are i.i.d random variables with μ being the mean
  - $P(|Y - \mu| > \epsilon) \to 0 \; as \; n \; \to \infty$
  - Limitations: Heavy tails (e.g., Cauchy distributions, Pareto distributions)

Examples

- Central limit theorem: Let $Y = \frac{1}{n}\sum_{i=1}^{n} X_i$, where X_i are i.i.d random variables with μ being the mean, σ^2 be the variance
  - $P\left(|Y-\mu| > \frac{\alpha}{\sqrt{n}}\right) \to 2\phi\left(-\frac{\alpha}{\sigma}\right) as\ n\ \to \infty,$ with ϕ being the standard norm CDF

Examples

- Probably Approximately Correct (PAC) learning



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Measurable Spaces

- (**S**, **B**) is a measurable space
  - **S** is a set
  - **B** is a σ-algebra on **S**, which is a collection of subsets of **S**
    - It contains $\emptyset$
    - Closed under complementation in **S**
    - Closed under countable union
  - The elements of **B** are called measurable sets

Measures: Definitions

- A signed (finite) measure on $(\boldsymbol{S}, \boldsymbol{B})$ is a countably additive map $\mu: \boldsymbol{B} \to \boldsymbol{R}$ such that $\mu(\emptyset) = 0$

- Positive signed measure: $\mu(A) \geq 0$ for all $A \in \boldsymbol{B}$

- A positive measure is a probability measure if $\mu(S) = 1$

- …is a subprobability measure if $\mu(S) \leq 1$

Measurable Functions

- $(\boldsymbol{S}, \boldsymbol{B_S})$ and $(\boldsymbol{T}, \boldsymbol{B_T})$ are measurable spaces. A function $f: \boldsymbol{S} \to \boldsymbol{T}$ is measurable if $f^{-1}(\boldsymbol{B}) = \{x \in S | f(x) \in B\}$ for every $\boldsymbol{B} \in \boldsymbol{B_T}$ is a measurable subset of S

Example: $$\chi_{\boldsymbol{B}}(s) = \begin{cases} 1, & s \in B, \\ 0, & s \notin B. \end{cases}$$

Measures: Lebesgue Integration

- **Lebesgue integration**: given $(\boldsymbol{S}, \boldsymbol{B})$, $\mu: \boldsymbol{B} \to \boldsymbol{R}$, $f: \boldsymbol{S} \to \boldsymbol{R}$, where $\mathrm{m} < f < M$

$$\int f \, d\mu = \lim_{n \to max} \sum_{i=0}^{n} f(s_i)\mu(B_i)$$

where $B_0, .., B_n$ is a measurable partition of $\boldsymbol{S}$, and the value of f does not vary more than $(M - m)/n$ in any B_i and $s_i \in B_i$

Probability Space

- Probability space (Ω, ε, P)
  - Ω is the sample space
  - ε is the $\sigma - algebra$ on Ω
  - P is a probability measure

- A random variable is a borel measurable function mapping each point in Ω to a real number uniquely
  - We shall call the space of random variables $\mathrm{X} \subseteq \mathrm{R}$

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Chebychev's Inequality

Theorem 1 *For some $a \in X$ where $X \subseteq R$, let f be a non-negative function such that $\{f(x) \geq b | \forall x \geq a\}$, where $b \in Y$ where $Y \subseteq R$. Then the following inequality holds,*

$$P(x \geq a) \leq \frac{E\{f(x)\}}{b}$$

Chebychev's Inequality:

$$P(|x - E(x)| \geq a) \leq \frac{E\{|x - E(x)|^2\}}{a^2} = \frac{Var\{x\}}{a^2}$$

Chebychev's Inequality

Chebychev's Inequality:

$$P(|x - E(x)| \geq a) \leq \frac{E\{|x - E(x)|^2\}}{a^2} = \frac{Var\{x\}}{a^2}$$

k	Min. % within k standard deviations of mean	Max. % beyond k standard deviations from mean
1	0%	100%
$\sqrt{2}$	50%	50%
1.5	55.55%	44.44%
2	75%	25%
$2\sqrt{2}$	87.5%	12.5%
3	88.8888%	11.1111%
4	93.75%	6.25%
5	96%	4%
6	97.2222%	2.7778%
7	97.9592%	2.0408%
8	98.4375%	1.5625%
9	98.7654%	1.2346%
10	99%	1%

https://en.wikipedia.org/wiki/Chebyshev%27s_inequality

Markov's Inequality

Theorem 1 *For some $a \in X$ where $X \subseteq R$, let f be a non-negative function such that $\{f(x) \geq b | \forall x \geq a\}$, where $b \in Y$ where $Y \subseteq R$. Then the following inequality holds,*

$$P(x \geq a) \leq \frac{E\{f(x)\}}{b}$$

Markov's Inequality:

$$P(x \geq a) \leq \frac{E\{x\}}{a}$$

Proof of Theorem 1

Theorem 1 *For some $a \in X$ where $X \subseteq R$, let f be a non-negative function such that $\{f(x) \geq b | \forall x \geq a\}$, where $b \in Y$ where $Y \subseteq R$. Then the following inequality holds,*

$$P(x \geq a) \leq \frac{E\{f(x)\}}{b}$$

Proof Sketch:

1. Let $X_1 = \{x \mid x \geq a \wedge x \in X\}$
2. $\int_X f dP \geq \int_{X_1} f dP \geq b \int_{X_1} dP$
3. Since $\int_X f dP = E\{f(x)\}, b \int_{X_1} dP = bP(x \geq a)$, the conclusion holds

Jensen's Inequality

Definition A function $\phi(x)$ is defined to be convex in interval (a, b) if for every point x' in the interval (a, b) there exists an m such that

$$\phi(x) \geq m(x - x') + \phi(x') \tag{4}$$

for any $x \in (a, b)$ ■

Theorem 2 *Let f and μ be measurable functions of x which are finite a.e. on $A \subseteq R^n$. Now let $f\mu$ and μ be integrable on A and $\mu \geq 0$. If ϕ is a function which is convex in interval (a, b) which is the range of function f and $\int_A \phi(f)\mu$ exists then,*

$$\phi\left(\frac{\int_A f\mu}{\int_A \mu}\right) \leq \frac{\int_A \phi(f)\mu}{\int_A \mu} \tag{5}$$

Jensen's Inequality

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Let μ be a probability measure, the we have:

$$E[\phi(f(x))] \geq \phi(E[f(x)])$$

Jensen's Inequality: Applications

$$E[\phi(f(x))] \geq \phi(E[f(x)])$$

Let $\phi(x) = -logx, f(x) = P(x)$, then we have the entropy function on a distribution:

$$Entropy(X) = -\int_X \log p(x)\, dp = -\int_X \log p(x)\, dp = E(-\log[p(x)]) \geq -logE(p(x))$$

$$\mathrm{H}(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Similarly, we can show KL divergence is always non-negative.

Jensen's Inequality: Proof

Theorem 2 *Let f and μ be measurable functions of x which are finite a.e. on $A \subseteq R^n$. Now let $f\mu$ and μ be integrable on A and $\mu \geq 0$. If ϕ is a function which is convex in interval (a, b) which is the range of function f and $\int_A \phi(f)\mu$ exists then,*

$$\phi\left(\frac{\int_A f\mu}{\int_A \mu}\right) \leq \frac{\int_A \phi(f)\mu}{\int_A \mu} \tag{5}$$

Proof Sketch:

1. Consider $x' = \frac{\int_A f\mu}{f_A \mu}$, then a $< x' < b$
2. Given ϕ is convex, we have $\phi\big(f(x)\big) \geq m(f(x) - x') + \phi(x')$
3. $\int_A \phi(f)\mu \geq m\left(\int_A f\mu - x' \int_A \mu\right) + \phi(x') \int_A \mu$
4. $\int_A \phi(f)\mu \geq \phi(x') \int_A \mu = \phi\left(\frac{\int_A f\mu}{f_A \mu}\right) \int_A \mu$

Hoeffding's Inequality

Theorem 5 *Let $x_1, \ldots, x_n$ be independent bounded random variables such that the random variable x_i falls in the interval $[p_i, q_i]$. Then for any $a > 0$ we have*

$$P(\sum_{i=1}^{n} x_i - E(\sum_{i=1}^{n} x_i) \geq a) \leq e^{-\frac{2a^2}{\sum_{i=1}^{n}(q_i - p_i)^2}}$$

Estimation of the mean: $P\left(\frac{\sum_{i=1}^{n} x^i}{n} - \mu \geq a\right) \leq e^{-\frac{2na^2}{(p-q)^2}}$

$$P\left(\left|\frac{\sum_{i=1}^{n} x^i}{n} - \mu\right| \geq a\right) \leq 2e^{-\frac{2na^2}{(p-q)^2}}$$

Hoeffding's Inequality

Estimation of the mean: $P\left(\frac{\sum_{i=1}^{n} x^i}{n} - \mu \geq a\right) \leq e^{-\frac{2na^2}{(p-q)^2}}$

$$P\left(\left|\frac{\sum_{i=1}^{n} x^i}{n} - \mu\right| \geq a\right) \leq 2e^{-\frac{2na^2}{(p-q)^2}}$$

a = 0.001

Sample Size (n)	failure Probability (δ)	Bounded Confidence (1−δ)
1	1.999996	0
10	1.999960	0
1,000	1.996004	0
10,000	1.960397	0
100,000	1.637462	0
1,000,000	0.270671	72.93%
10,000,000	~$4 * 10^{-9}$	100.00%

Hoeffding's Inequality: Proof

Chernoff's Bound $P(x \geq a) \leq \frac{Ee^{sx}}{e^{sa}}$

Theorem 5 *Let $x_1, \ldots, x_n$ be independent bounded random variables such that the random variable x_i falls in the interval $[p_i, q_i]$. Then for any $a > 0$ we have*

$$P(\sum_{i=1}^{n} x_i - E(\sum_{i=1}^{n} x_i) \geq a) \leq e^{-\frac{2a^2}{\sum_{i=1}^{n}(q_i - p_i)^2}}$$

Proof Sketch:

1. $P(x - E(x) \geq a) \leq \frac{Ee^{s(x-E(x))}}{e^{sa}}$
2. $P(S_n - E(S_n) \geq a) \leq \frac{Ee^{s(S_n - E(S_n))}}{e^{sa}} \leq e^{-sa} \prod_{i=1}^{n} Ee^{s(x_i - E(x_i))}$, where $S_n = \sum_{i=1}^{n} x_i$
3. Let y be any random variable where $p \leq y \leq q$, and $Ey = 0$. Then for any s>0:
$$e^{sy} \leq \frac{y-p}{q-p} e^{sq} + \frac{q-y}{q-p} e^{sp}$$
$$Ee^{sy} \leq \frac{-p}{q-p} e^{sq} + \frac{q}{q-p} e^{sp}$$
4. Let $\alpha = \frac{-p}{p-q}$, then $Ee^{sy} \leq \alpha e^{sq} + (1-\alpha)e^{sp} = \left(\alpha e^{s(q-p)} + (1-\alpha)\right) e^{-s\alpha(q-p)} = e^{\log\left(\alpha e^{s(q-p)} + (1-\alpha)\right) - s\alpha(q-p)}$
5. $Ee^{sy} \leq e^{\phi(u)}$, where $u = s(q-p), \phi(u) = \log\left(\alpha e^{u} + (1-\alpha)\right) - \alpha u$

Hoeffding's Inequality: Proof

Proof Sketch:

5. $Ee^{sy} \leq e^{\phi(u)}$, where $u = s(q-p), \phi(u) = \log\big(\alpha e^{u} + (1-\alpha)\big) - \alpha u$
6. $\phi(u) = \phi(0) + x\phi'(0) + \frac{x^2}{2}\phi''(\eta)$ for some η
7. We have $\phi(0) = 0, \phi'(0) = 0, \phi''(x) = \frac{\alpha(1-\alpha)e^{x}}{(1-\alpha+\alpha e^{x})^2}$, $\max \phi'' = \frac{1}{4}$, when $e^{x} = \frac{1-\alpha}{\alpha}$
8. $Ee^{sy} \leq e^{\frac{u^2}{8}} = e^{\frac{s^2(q-p)^2}{8}}$
9. $P(S_n - E(S_n) \geq a) \leq e^{-sa}\prod_{i=1}^{n} Ee^{s(x_i - E(x_i))} \leq e^{-sa}\prod_{i=1}^{n} e^{\frac{s^2(q_i-p_i)^2}{8}} = e^{-sa}e^{\sum_{i=1}^{n}\frac{s^2(q_i-p_i)^2}{8}}$
10. $min\ e^{-sa}e^{\sum_{i=1}^{n}\frac{s^2(q_i-p_i)^2}{8}} = e^{-\frac{2a^2}{\sum_{i=1}^{n}(q_i-p_i)^2}}$

Hoeffding's Inequality: Fairness Application

- Group parity: $\frac{P(accept \mid minor)}{P(accept \mid major)} > \eta$

- Equal opportunity: $\frac{P(accept \mid minor \wedge qualified)}{P(accept \mid major \wedge qualified)} > \eta$

Hoeffding's Inequality: Adding Samples

- With the current number of samples, I cannot get a tight bound. Can I keep adding samples until the bound is satisfactory?

- Adaptive Concentration Inequalities for Sequential Decision Problems. NIPS'16.

Summary

- Concentration equality: theoretical guarantees on estimating the expectation using samples

- Chebychev's Inequality
- Jensen's Inequality
- Hoeffding's Inequality